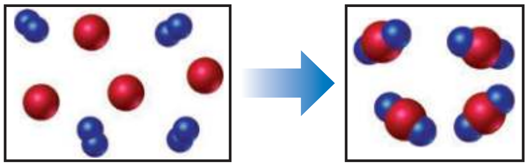
Exercises

Visualizing Concepts

3.61 The reaction between reactant A (blue spheres) and reactant B (red spheres) is shown in the following diagram:



Based on this diagram, which equation best describes the reaction? [Section 3.1]

(**a**) $A_2 + B \longrightarrow A_2B$

(**b**) $A_2 + 4\,B \longrightarrow 2\,AB_2$

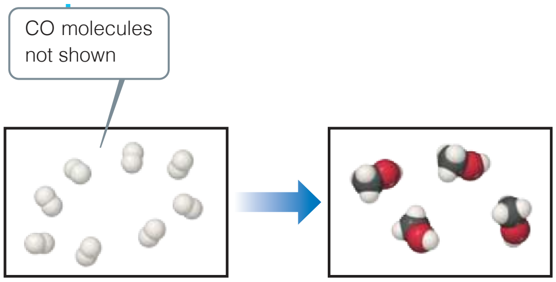
(**c**) $2\,A + B_4 \longrightarrow 2\,AB_2$

(**d**) $A + B_2 \longrightarrow AB_2$

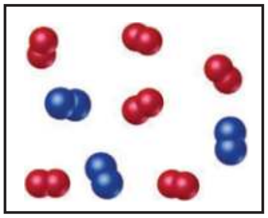
3.62 The following diagram shows the combination reaction between hydrogen, H_2, and carbon monoxide, CO, to produce methanol, CH_3OH (white spheres are H, black spheres are C, red spheres are O). The correct number of CO molecules involved in this reaction is not shown. [Section 3.1]

(**a**) Determine the number of CO molecules that should be shown in the left (reactants) box.

(**b**) Write a balanced chemical equation for the reaction.



3.63 The following diagram represents the collection of elements formed by a decomposition reaction. (**a**) If the blue spheres represent N atoms and the red ones represent O atoms, what was the empirical formula of the original compound? (**b**) Could you draw a diagram representing the molecules of the compound that had been decomposed? Why or why not? [Section 3.2]



3.64 The following diagram represents the collection of CO_2 and H_2O molecules formed by complete combustion of a hydrocarbon. What is the empirical formula of the hydrocarbon? [Section 3.2]

3.65 Glycine, an amino acid used by organisms to make proteins, is represented by the following molecular model.

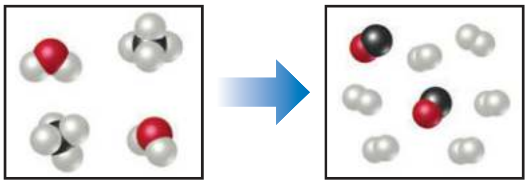
(**a**) Write its molecular formula.

(**b**) Determine its molar mass.

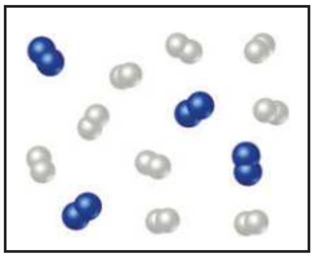
(**c**) Calculate how many moles of glycine are in a 100.0-g sample of glycine.

(**d**) Calculate the percent nitrogen by mass in glycine. [Sections 3.3 and 3.5]

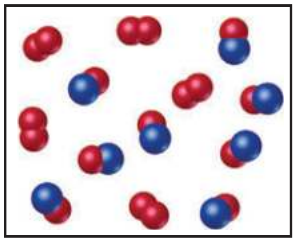
3.66 The following diagram represents a high-temperature reaction between CH_4 and H_2O. Based on this reaction, find how many moles of each product can be obtained starting with 4.0 mol CH_4. [Section 3.6]



3.67 Nitrogen (N_2) and hydrogen (H_2) react to form ammonia (NH_3). Consider the mixture of N_2 and H_2 shown in the accompanying diagram. The blue spheres represent N, and the white ones represent H. (**a**) Write the balanced chemical equation for the reaction. (**b**) What is the limiting reactant? (**c**) How many molecules of ammonia can be made, assuming the reaction goes to completion, based on the diagram? (**d**) Are any reactant molecules left over, based on the diagram? If so, how many of which type are left over? [Section 3.7]



3.68 Nitrogen monoxide and oxygen react to form nitrogen dioxide. Consider the mixture of NO and O_2 shown in the accompanying diagram. The blue spheres represent N, and the red ones represent O. (**a**) How many molecules of NO_2 can be formed, assuming the reaction goes to completion? (**b**) What is the limiting reactant? (**c**) If the actual yield of the reaction was 75% instead of 100%, how many molecules of each kind would be present after the reaction was over?



Chemical Equations and Simple Patterns of Chemical Reactivity (Section 3.1)

3.69 A key step in balancing chemical equations is correctly identifying the formulas of the reactants and products. For example, consider the reaction between calcium oxide, $CaO(s)$, and $H_2O(l)$ to form aqueous calcium hydroxide. (**a**) Write a balanced chemical equation for this combination reaction, having correctly identified the product as $Ca(OH)_2(aq)$. (**b**) Is it possible to balance the equation if you incorrectly identify the product as $CaOH(aq)$, and if so, what is the equation?

3.70 Balance the following equations:

(**a**) $HClO_4(aq) + P_4O_{10}(s) \longrightarrow HPO_3(aq) + Cl_2O_7(l)$

(**b**) $Au_2S_3(s) + H_2(g) \longrightarrow Au(s) + H_2S(g)$

(**c**) $Ba_3N_2(s) + H_2O(aq) \longrightarrow Ba(OH)_2(aq) + NH_3(g)$

(**d**) $Na_2CO_3(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$

3.71 Balance the following equations:

(**a**) $CF_4(l) + Br_2(g) \longrightarrow CBr_4(l) + F_2(g)$

(**b**) $Cu(s) + HNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + NO_2(g) + H_2O(l)$

(**c**) $MnO_2(s) + HCl(aq) \longrightarrow MnCl_2(s) + H_2O(l) + Cl_2(g)$

(**d**) $KOH(aq) + H_3PO_4(aq) \longrightarrow K_3PO_4(aq) + H_2O(l)$

3.72 Write balanced chemical equations to correspond to each of the following descriptions: (**a**) When sulfur trioxide gas reacts with water, a solution of sulfuric acid forms.(**b**) Boron sulfide, $B_2S_3(s)$, reacts violently with water to form dissolved boric acid, H_3BO_3, and hydrogen sulfide gas. (**c**) Phosphine, $PH_3(g)$, combusts in oxygen gas to form water vapor and solid tetraphosphorus decaoxide. (**d**) When solid mercury(II) nitrate is heated, it decomposes to form solid mercury(II) oxide, gaseous nitrogen dioxide, and oxygen. (**e**) Copper metal reacts with hot concentrated sulfuric acid solution to form aqueous copper(II) sulfate, sulfur dioxide gas, and water.

Patterns of Chemical Reactivity (Section 3.2)

3.73 (**a**) When a compound containing C, H, and O is completely combusted in air, what reactant besides the hydrocarbon is involved in the reaction? (**b**) What products form in this reaction? (**c**) What is the sum of the coefficients in the balanced chemical equation for the combustion of one mole of acetone, $C_3H_6O(l)$, in air?

3.74 Write a balanced chemical equation for the reaction that occurs when (**a**) titanium metal reacts with $O_2(g)$; (**b**) silver(I) oxide decomposes into silver metal and oxygen gas when heated; (**c**) propanol, $C_3H_7OH(l)$ burns in air; (**d**) methyl *tert*-butyl ether, $C_5H_{12}O(l)$, burns in air.

3.75 Balance the following equations and indicate whether they are combination, decomposition, or combustion reactions:

(**a**) $NaClO_3(s) \longrightarrow NaCl(s) + O_2(g)$

(**b**) $NH_4OH(aq) \longrightarrow H_2O(l) + NH_3(g)$

(**c**) $K(s) + Cl_2(g) \longrightarrow KCl(s)$

(**d**) $C_2H_5OH(l) + O_2(g) \longrightarrow CO_2(g) + H_2O(l)$

Formula Weights (Section 3.3)

3.76 Determine the formula weights of each of the following compounds: (**a**) Butyric acid, $CH_3CH_2CH_2COOH$, which is responsible for the rotten smell of spoiled food; (**b**) sodium perborate, $NaBO_3$, a substance used as bleach; (**c**) calcium carbonate, $CaCO_3$, a substance found in marble. (c) CF_2Cl_2, a refrigerant known as Freon; (**d**) $NaHCO_3$, known as baking soda and used in bread and pastry baking; (**e**) iron pyrite, FeS_2, which has a golden appearance and is known as "Fool's Gold."

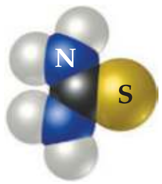
3.77 Calculate the percentage by mass of the indicated element in the following compounds: (**a**) hydrogen in methane, CH_4, the major hydrocarbon in natural gas; (**b**) oxygen in vitamin E, $C_{29}H_{50}O_2$; (**c**) sulphur in magnesium sulphate, $MgSO_4$, a substance used as a drying agent; (**d**) nitrogen in epinephrine, $C_9H_{13}NO_3$, also known as adrenalin, a hormone that is important for the fight-or-flight response; (**e**) oxygen in the insect pheromone sulcatol, $C_8H_{16}O$; (**f**) carbon in sucrose, $C_{12}H_{22}O_{11}$, the compound that is responsible for the sweet taste of table sugar.

3.78 Calculate the percentage of carbon by mass in each of the compounds represented by the following models:

(**a**)

(**b**)

(**c**)



(**d**)

Avogadro's Number and the Mole (Section 3.4)

3.79 (**a**) What is the mass, in grams, of one mole of ^{79}Br? (b) How many bromine atoms are present in one mole of ^{79}Br?

3.80 Without doing any detailed calculations (but using a periodic table to give atomic weights), rank the following samples in order of increasing numbers of atoms: 0.2 mol PCl_5 molecules, 80 g Fe_2O_3, 3.0×10^{23} CO molecules.

3.81 Calculate the following quantities:

(**a**) mass, in grams, of 1.50×10^{-2} mol CdS

(**b**) number of moles of NH_4Cl in 86.6 g of this substance

(**c**) number of molecules in 8.447×10^{-2} mol C_6H_6

(**d**) number of O atoms in 6.25×10^{-3} mol $Al(NO_3)_3$

3.82 (**a**) What is the mass, in grams, of 1.223 mol of iron(III) sulfate?

(**b**) How many moles of ammonium ions are in 6.955 g of ammonium carbonate?